

FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility
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STANDARD OPERATING PROCEDURE

Quality Inspection Procedure for Water Treatment Line 90

Department: Water Treatment

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Approved by: Plant Manager

Document Control Information

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Title	Quality Inspection Procedure for Water Treatment Line 9
Department	Water Treatment
Plant	FreshFlow Bottling Plant - Pune Production Facility
Prepared By	Employee 2
Reviewed By	Anita Desai
Approved By	Rajesh Kulkarni - Plant Manager
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1. Purpose

Produce water meeting the FreshFlow Beverages water quality specification (TDS <50 ppm, pH 6.5-7.5, microbiologically safe) for use in product production and CIP.

2. Scope

Water Treatment operators and Maintenance team.

3. Prerequisites and Requirements

3.1 Training Requirements

- RO system operation training
- Chemical dosing training

3.2 Tools and Materials Required

- Calibrated instruments as listed in the plant calibration schedule.
- Appropriate PPE as specified in Section 4.
- Completed permit-to-work (if required for maintenance activities).
- Access to relevant equipment manual and P&ID drawings.
- CMMS access to create and close work orders.

4. Health, Safety and Environmental Requirements

! WARNING

This SOP must not be carried out without appropriate PPE and, where required, a valid Permit to Work (SOP-SAF-006). If in doubt about safety, STOP and consult the Safety Officer (EMP009 / EMP017).

4.1 Identified Hazards

- High-pressure RO system (up to 12 bar) - pressure release risk
- UV radiation from UV sterilizer lamps - eye and skin damage risk
- Chemical dosing agents (chlorine, antiscalant, acid) - chemical hazard

4.2 Emergency Response

In case of any emergency during this procedure: Press the nearest Emergency Stop, alert personnel in the vicinity, call the Safety Officer and Shift Supervisor. Refer to the Emergency Response Plan (SOP-SAF-004) and site evacuation procedure.

5. Detailed Procedure

5.1 Pre-Task Verification

1. Confirm the relevant equipment is available and no conflicting activities are planned.
2. Check shift handover log for any outstanding issues with this equipment.
3. Verify all required materials, tools, and chemicals are available and correctly labelled.
4. Obtain all required permits (PTW, hot work, confined space as applicable).
5. Brief all involved personnel on this SOP before starting.

5.2 Main Procedure

1. Isolate and LOTO the equipment per SOP-SAF-002. Verify zero energy state.
2. Allow equipment to cool to ambient temperature before touching internal parts.
3. Disassemble relevant sub-assembly according to the equipment manual section 6.
4. Inspect all removed parts. Photograph any damaged or worn components.
5. Clean the housing and contact surfaces. Remove all old gasket material.
6. Install new parts using food-grade lubricants where applicable. Do not over-torque fasteners.
7. Reassemble in reverse order. Verify all fasteners are tightened to specified torques.
8. Remove LOTO and test equipment at low speed/load before returning to production.
9. Verify key parameters (pressure, temperature, vibration) after restart.
10. Record the work in CMMS with parts used, cost, and technician details.